

Tutorial-2, Probability and Statistics (MCC505)
Monsoon Semester: 20-21

1. If A is an event, then the conditional probability of A given A is
 - a) 1
 - b) 0
 - c) infinity

2. If $B \subset A$, then $P(A/B) = ?$
 - a) 1
 - b) 0
 - c) $\frac{P(A)}{P(B)}$
 - d) $\frac{P(B)}{P(A)}$

3. With a pair of dice thrown at a time, the probability of getting a sum more than 9 is
 - a) 5/6
 - b) 7/36
 - c) 1/6
 - d) none of the above.

4. The distribution function of a random variable X is
$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b. \end{cases}$$
Find $E(X)$ and $\text{Var}(X)$.

5. If distribution function of a random variable is $F(x) = 1 - e^{-2x}$, for $x > 0$, find $P(2 \leq X \leq 5)$.

6. If joint pdf of (X, Y) is $f(x, y) = x^2 + \frac{xy}{3}$ for $0 < x < 1$, $0 < y < 2$. Find $P(Y \leq \frac{1}{2})$.

7. Suppose that X is a random variable that takes on values 0, 2 and 3 with probabilities 0.3, 0.1, 0.6 respectively. Let $Y = 3(X - 1)^2$.
(a) What is the expectation of X? (b) What is the expectation of Y?

8. Suppose that the cdf of X is given by: $F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } 0 \leq x < 2 \\ 2 & \text{for } 2 \leq x < 4 \\ 1 & \text{for } x \geq 4. \end{cases}$

Determine the pmf of X.

9. Let X be a random variable with cdf $F(x) = 2x^2 - x^4$, for $0 \leq x \leq 1$.

(a) Compute $P(\frac{1}{4}X \leq \frac{3}{4})$.

(b) What is the pdf of X?

10.

	Y=1	Y=2	Y=3	
X=1	$\frac{1}{6}$	0	...	$\frac{1}{3}$
X= 2	...	$\frac{1}{4}$	...	$\frac{1}{3}$
X=3	...	...	$\frac{1}{4}$	...
	$\frac{1}{6}$	$\frac{1}{3}$	...	1

Complete the table.